DESIGN REPORT

FITNESS MEMBERSHIP MANAGEMENT SYSTEM

SESEIU24001 - Võ Châu Anh

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1. Assumptions

The problem statement is open, so these are the decisions I made:

- Currency is VND, stored as `Double` and formatted by a `vnd` helper.
- `id` and `activationDate` are immutable (`let`) — identity never changes after registration.
- Three membership plans: Basic, Premium, Personal Training. Each has its own facilities, hours, and fee model.
- Three user roles: `member`, `staff`, `manager`. Member can only use facilities and submit change requests; staff handles day-to-day operations; manager additionally controls billing amounts and can expire/reactivate memberships.
- Monthly billing period. `billingStart` records the current period's start; `startNewBillingPeriod` resets usage counters and requires active status + zero balance.
- Temporary upgrade is only available on Basic (1…30 days), and grants Premium-level facilities and hours while active. Only one upgrade at a time.
- Promotions are pre-seeded (`Promotion.samples`) and can only be applied if valid, eligible for the plan, and not already applied.
- Check-in requires: membership usable (not suspended/expired), facility included in the plan, and time inside the access window.
- Change requests (freeze, unfreeze, trial-Premium) are submitted by the member and resolved by staff/manager. Only one pending request at a time.
- No backend — `MembershipManager(sample: true)` loads six members; `MembershipManager.preview()` reuses it for SwiftUI previews.

2. Type identification

Classes:

Class	Responsibility
`Membership`	Base class. Holds identity (`id`, `activationDate`), shared state (`memberName`, `phone`, `status`, `unitFee`, `amountPaid`, `promotion`, `temporaryUpgradeUntil`, `pendingRequest`, `checkIns`, `billingStart`). Owns all shared workflow: payment recording, promotion apply/remove, temporary upgrade, status change, check-in, change requests, billing period reset, and the shared `authorize`/`ensureUsable` guards.

BasicMembership	Plan-specific overrides only. Provides base facilities (gym, cardio, locker) and a 6–21 access window. Supports temporary upgrade — while active, uses PremiumMembership.perkFacilities and PremiumMembership.perkWindow. Has no usage counter, so it inherits the base no-op resetPeriodUsage().
PremiumMembership	Overrides facilities (all areas incl. pool, sauna, group studio), a wider 5–23 window, and adds group-class booking with a monthly limit. Overrides resetPeriodUsage() to clear classesUsed and bookedClasses.
PersonalTrainingMembership	Session-based plan. Tracks packageSessions, addedSessions, sessionsAttended, and enforces "sessions cannot go negative". Overrides resetPeriodUsage() to clear addedSessions and sessionsAttended.
MembershipManager	ObservableObject. Owns [Membership] and promotions (Promotion.samples). Provides search, add, aggregate totals, and the attempt(_:) wrapper that turns thrown errors into visible messages.

Structs:

Struct	Fields	Purpose
AccessWindow	let openHour: Int, let closeHour: Int; computed text, contains(_:)	Opening hours policy used for check-in validation.
Promotion	id (from code), code, title, percentOff, eligible: Set<MembershipType>, validUntil; methods isValid(on:), isEligible(for:); static samples	Discount configuration.
ChangeRequest	let id = UUID(), kind, date	One pending member request.
CheckInRecord	let id = UUID(), date, facility	One entry record.
ChargeLine	let id = UUID(), label, amount	One row on a bill.
ChargeBreakdown	lines, promotion, subtotal, discount, total, paid, outstanding	Frozen snapshot produced by the shared breakdown computed property.

InfoItem	let id = UUID(), label, value	Generic display row.

Supporting enums: MembershipType (basic/premium/personalTraining), MembershipStatus (active/suspended/expired), Facility (7 values with an icon), Permission (8 cases), UserRole (member/staff/manager, with can(_:)), RequestKind (freeze/unfreeze/trialPremium). MembershipError covers every validation path currently used.

3. Properties & Methods

Shared stored properties in Membership:

- let id, let activationDate — immutable
- private(set) var memberName, phone, status, unitFee, amountPaid, promotion, temporaryUpgradeUntil, pendingRequest, checkIns: [CheckInRecord], billingStart

Overridable policy hooks in Membership:

- type, typeName, icon, feeLabel, facilities, accessWindow, supportsTemporaryUpgrade, benefitRows, searchableText
- chargeLines() -> [ChargeLine] — main polymorphic fee calculator
- resetPeriodUsage() — no-op in base; overridden by the subclasses that track usage

Shared final members (cannot be overridden):

- breakdown: ChargeBreakdown (computed), estimatedCharges, outstanding
- authorize(_:_:action:), ensureUsable()
- updateDetails(name:phone:by:), setStatus(_:by:), updateFee(to:by:), recordPayment(_:by:)
- applyPromotion(_:by:), removePromotion(by:), grantTemporaryUpgrade(days:by:)
- startNewBillingPeriod(by:), checkIn(at:facility:by:)
- submitRequest(_:by:), resolveRequest(approve:by:)

Shared non-final members that are not overridden by any current subclass:

- hasTemporaryUpgrade, activePromotion, detailRows

Shared private helper:

- applyUpgrade(days:) — called from both grantTemporaryUpgrade and resolveRequest(.trialPremium) so the upgrade rules live in one place

Subclass-specific properties:

Class	Property	Access
BasicMembership	static baseFacilities, static baseWindow	let (constants)

PremiumMembership	static perkFacilities, static perkWindow	let (constants)
PremiumMembership	monthlyClassLimit: Int	let
PremiumMembership	classesUsed, bookedClasses: [String]	private(set) var
PersonalTrainingMembership	packageSessions: Int	let
PersonalTrainingMembership	coachName, addedSessions, sessionsAttended	private(set) var

Subclass-specific computed properties: PremiumMembership.classesRemaining, PersonalTrainingMembership.sessionsPurchased / sessionsRemaining.

Subclass-specific methods:

- PremiumMembership.bookGroupClass(named:by:) — enforces the monthly limit
- PersonalTrainingMembership.attendSession(by:) — rejects when sessionsRemaining == 0
- PersonalTrainingMembership.addSessions(_:by:) — accepts only 1…20
- PremiumMembership.resetPeriodUsage() and PersonalTrainingMembership.resetPeriodUsage() override the base no-op; BasicMembership keeps the no-op because it tracks no periodic usage.

4. Inheritance & Polymorphism

Hierarchy:

```
Membership
├── BasicMembership
├── PremiumMembership
└── PersonalTrainingMembership
```

Membership is not final (it must be inheritable). The three subclasses are final, so no deeper extension is created by accident. All shared lifecycle methods are final in the base, so subclasses can only override the policy hooks — this keeps billing math, promotion handling, and status transitions consistent across plans.

Inherited (written once): identity, payment recording, promotion apply/remove, temporary upgrade, status change with authorization, check-in workflow, change request workflow, breakdown and outstanding calculation.

Complete override matrix:

Member	BasicMembership	PremiumMembership	PersonalTrainingMembership

type	.basic	.premium	.personalTraining
typeName	"Basic" / "Basic (+ upgrade)"	"Premium"	"Personal Training"
icon	figure.walk	star.circle.fill	figure.strengthtraining.traditional
feeLabel	—	—	"Price per session"
facilities	3 base areas; Premium set while upgraded	6 perk areas (incl. pool, sauna, studio)	4 areas (gym, cardio, locker, training zone)
accessWindow	06:00–21:00; 05:00–23:00 while upgraded	05:00–23:00 (perk)	—
supportsTemporaryUpgrade	true	—	—
benefitRows	access + hours + optional upgrade line	access + hours + classes-used	coach + sessions-remaining + attended + hours
chargeLines()	Monthly fee = unitFee	Monthly fee = unitFee	Package: packageSessions × unitFee + Added: addedSessions × unitFee
resetPeriodUsage()	—	clears classesUsed, bookedClasses	clears addedSessions, sessionsAttended
searchableText	—	joined bookedClasses	coachName

A dash (—) means the subclass does not override that member and uses the base default. `detailRows` is defined once in `Membership` and is not overridden — plan-specific info is delivered through `benefitRows` and `chargeLines()`.

Polymorphism in practice:

- `MembershipManager.totalEstimated` sums `estimatedCharges` over every `Membership`. `estimatedCharges` uses the shared `breakdown`, which calls `chargeLines()` — each plan provides its own fee decomposition, and the manager never branches on type.

- `BillCard` and `MemberDetailView` render `breakdown.lines` and `benefitRows` generically. The only type casts are in `MemberDetailView` to attach `PremiumSection` / `TrainingSection` action forms, and in `UpgradeSection` which is conditionally shown for plans where `supportsTemporaryUpgrade == true` (only Basic today, but future plans can opt in).
- Cross-subclass dependency: `BasicMembership` uses `PremiumMembership.perkFacilities` / `PremiumMembership.perkWindow` when upgraded. This is a deliberate reuse of the Premium constants rather than duplicating them, at the cost of a small structural coupling between two concrete subclasses.

When protocol/composition would be better: if future plans can combine features (e.g., a PT plan that also includes pool access), a class tree starts to break. Protocols like `Payable`, `Bookable`, `CheckInEnabled` combined with small value types (a `UsageQuota`, a `Perks` struct) would then be cleaner. I chose class inheritance here because every plan shares the same lifecycle (`breakdown`, `recordPayment`, `checkIn`, request/approve), and the differences are entirely within the fee formula, facilities set, access window, and usage counter — all expressible as overrides.

5. Encapsulation & Validation

Access-control decisions:

Data	Access modifier	Reason
`Membership.id`, `activationDate`	`let`	Identity — must never change
`Membership.memberName`, `phone`	`private(set) var`	Only updated by `updateDetails` with trim + non-empty validation
`Membership.status`, `unitFee`, `amountPaid`, `promotion`, `temporaryUpgradeUntil`, `pendingRequest`, `billingStart`	`private(set) var`	Controlled state — changed only through a small well-defined set of mutating methods, each of which is authorized and validated first
`Membership.checkIns`	`private(set) var`	Only appended by `checkIn`
`PremiumMembership.monthlyClassLimit`,	`let`	Fixed at creation

PersonalTrainingMembership.packageSessions		
PremiumMembership.classesUsed, bookedClasses	private(set) var	Only mutated by bookGroupClass / resetPeriodUsage
PersonalTrainingMembership.coachName, addedSessions, sessionsAttended	private(set) var	Only mutated by their dedicated methods
MembershipManager.memberships	@Published private(set)	Views read; only the manager appends
MembershipManager.currentRole	@Published (writable)	Selected by the user
MembershipManager.promotions	let = Promotion.samples	Seed catalogue; not mutated

Controlled mutators that write private(set) state: setStatus and resolveRequest are the only writers of status; recordPayment and startNewBillingPeriod are the only writers of amountPaid; applyPromotion / removePromotion are the only writers of promotion; submitRequest / resolveRequest are the only writers of pendingRequest; updateFee is the sole writer of unitFee; grantTemporaryUpgrade / resolveRequest (via the private applyUpgrade) are the sole writers of temporaryUpgradeUntil; startNewBillingPeriod is the sole writer of billingStart. Every one of these methods is final and begins with authorize(role, permission, action:) before any other logic runs.

Shared guard: ensureUsable() is used by every operation that requires an active membership (checkIn, attendSession, addSessions, applyUpgrade). Suspended and expired plans are rejected through one function.

Validation rules (per method):

Rule	Where	Error
Name/phone not empty (after trim)	updateDetails	emptyField
New status ≠ current status	setStatus	statusUnchanged
Expire/reactivate requires manageBilling	setStatus	unauthorized

Reactivate from expired requires zero balance	`setStatus`	`cannotReactivateWithBalance`
Fee finite and > 0	`updateFee`	`invalidAmount`
Payment finite and > 0	`recordPayment`	`invalidAmount`
Payment ≤ outstanding	`recordPayment`	`overpayment`
Promotion valid + eligible + not already applied	`applyPromotion`	`promotionExpired` / `promotionNotEligible` / `promotionAlreadyApplied`
Upgrade days 1…30, plan supports it, usable, no active upgrade	`applyUpgrade`	`invalidUpgradeDays` / `upgradeNotAvailable` / `upgradeAlreadyActive`
Billing period reset: active + zero outstanding	`startNewBillingPeriod`	`periodRequiresActive` / `outstandingBalance`
Check-in: usable + facility included + inside window	`checkIn`	`membershipSuspended` / `membershipExpired` / `facilityNotIncluded` / `outsideAccessHours`
Change request: no other pending + applicable kind	`submitRequest`	`requestPending` / `requestNotApplicable` / `upgradeNotAvailable` / `upgradeAlreadyActive`
Resolve request: requires one to exist	`resolveRequest`	`noPendingRequest`
Group class: name non-empty, limit not reached	`bookGroupClass`	`emptyField` / `monthlyLimitReached`
Attend session: balance > 0	`attendSession`	`insufficientSessions`

Add sessions: 1…20	addSessions	invalidSessionCount
No duplicate ID	addMembership	duplicateMembership
Role must have permission	authorize	unauthorized

6. SwiftUI Implementation

Screen structure:

- ContentView — TabView with Members and Billing. Injects MembershipManager(sample: true) as @StateObject.
- MembersView — search field, segmented filter by MembershipType, RoleBanner, and a list of MembershipCard. Cards show name, ID chip, plan pill, status pill, opening hours, and estimated charges + paid/due badge.
- BillingView — outstanding-balance banner, list of BillCard for every membership (line items + promotion discount + total/paid/outstanding).
- MemberDetailView — one shared layout for all plans:
    - Header (icon, name, ID, status pill) + acting-role row
    - Details (shared detailRows) + Edit button (EditMemberView sheet)
    - Plan benefits (polymorphic benefitRows, facilities as chips via FlowLayout)
    - Estimated charges (polymorphic breakdown.lines + promotion row + total/paid/outstanding)
    - Type-specific action forms: PremiumSection (group classes) or TrainingSection (sessions) — only the cast differs, the payslip stays polymorphic
    - UpgradeSection — shown only when supportsTemporaryUpgrade
    - CheckInSection, BillingSection (pay, change fee, apply/remove promo, start new period), StatusSection (change status + manage change requests)
- EditMemberView — sheet for name/phone only. ID and activation date are shown as read-only LabeledContent, matching the let immutability in code.

Where polymorphic behavior is visible:

- Plan cards and detail sections render different fee lines per type (Basic/Premium show one monthly-fee line; PT shows two lines when sessions were added) without any switch in the view — it iterates breakdown.lines.
- Benefit rows differ per plan without a switch — from benefitRows.
- Facilities and access window change automatically. After a temporary upgrade on Basic, the facilities chips switch to Premium areas and the card reflects it via typeName = "Basic (+ upgrade)".

- Group-class and coaching controls appear only on the relevant plan (type cast). The payslip renderer above them is fully polymorphic.

7. Conclusion

The design puts shared lifecycle behavior (payment, promotion, upgrade, check-in, request/approve, status transitions, breakdown math) in `Membership`, and pushes only plan-specific variance into subclasses through eleven overridable members: `type`, `typeName`, `icon`, `feeLabel`, `facilities`, `accessWindow`, `supportsTemporaryUpgrade`, `benefitRows`, `chargeLines`, `resetPeriodUsage`, and `searchableText` — each overridden by exactly the subclass that needs a different value, and inherited by the rest. `BasicMembership` keeps the base no-op `resetPeriodUsage()` because it has no periodic usage to clear; `PremiumMembership` and `PersonalTrainingMembership` override it to reset their own counters. Sensitive state (`id`, `activationDate`, `status`, `amountPaid`, `promotion`, `unitFee`, all usage counters) is `let` or `private(set)`, and each piece of that state has a small well-defined set of `final` mutating methods that all begin with `authorize(...)` then validate. Rejected operations are explicit (`MembershipError` covers every validation path currently used) and the app surfaces them as alert messages. The result is a compact, extensible model that is easy to demonstrate in SwiftUI with clear valid/invalid feedback.